Contact

+40 760 827 250 (Mobile)
filipescurobert@gmail.com

www.linkedin.com/in/robertfilipescu (LinkedIn)

Top Skills

Embedded C
Git/Gitlab
Microsoft Visual Studio Code
AUTOSAR Classic Platform
CANoe
Jira
INCA
Lauterbach TRACE32

Languages

Romanian (Native)
English (Professional Working)
German (Beginner)

Certifications

The Bits and Bytes of Computer Networking
Responsive Web Design
Technical Support Fundamentals
Vector Certified Embedded Associate

Robert Buzabrici-Filipescu

Senior Embedded Software Engineer at Capgemini Engineering
Iaşi, Romania

Summary

Contact information : Mobile : + 40 760 827 250
Email: filipescurobert@gmail.com

Senior Embedded Software Engineer with over 13 years of experience in automotive and space-grade systems, specializing in **instrument cluster development, CAN communication** and **AUTOSAR-based systems**. Proven track record of delivering robust and high-performance embedded solutions for OEMs including Daimler, Ford, Renault, Volkswagen, Porsche, and Audi.

- Reduced debugging time by 25% through rigorous code reviews and adherence to coding standards.
- Enhanced code reliability by 25% via comprehensive unit and component testing strategies.
- Delivered over 300 code commits, ensuring consistent quality and timely releases.
- Created a centralized knowledge base of 50+ software modules, reducing integration errors by 20% and improving cross-team collaboration.

Skilled in **ANSI C, CANoe, Trace32**, and **Google Test**, with hands-on experience in **V-Cycle** and **Agile development**, **diagnostics and calibration protocols (CCP, XCP)**, and **ISO 26262** compliance. Currently focused on expanding expertise in modern programming languages including **C++, C#, Python**, and **Rust** to drive innovation in embedded systems.
Seeking opportunities in collaborative environments that value technical excellence, mentorship, and continuous improvement.

Specialties:
Proficient: C, GIT, Gitlab
Intermediate: C++/C#, , HTML, JavaScript, CSS, MySQL, PostgreSQL, PVCS, Canoe/Canalyzer, MKS, IMS, RTC, Trace32, Multi, Jenkins, Linux
Basic: Java, PHP, Oracle, Doors, ESI, IMES, CCP/XCP

Experience

CAPGEMINI Engineering

Senior Embedded Software Engineer / Tech Lead Replacement

April 2022 – November 2025 (3 years 7 months)

Iaşi, Romania

- Spearheaded performance optimization of embedded modules, reducing memory usage by 20% and improving system responsiveness, which enabled smoother integration with OEM systems.
- Facilitated cross-functional collaboration, decreasing issue resolution time by 30%.
- Enhanced test coverage and reliability by refining CAPL scripts and test campaigns, leading to a 30% reduction in post-release bugs.
- Resolved issues across Mode Management and Body teams.
- Oversaw embedded software functionality on bench setups using Trace32 and CANoe, contributing to successful system integration.
- Reviewed and delivered 300+ high-quality code commits via Git, ensuring 100% adherence to established coding standards and project timelines, which led to a 25% reduction in debugging time for subsequent releases.
- Orchestrated comprehensive unit and component testing strategies utilizing Google Test, leading to a demonstrable 25% enhancement in code dependability and mitigating regression defects during critical system integration.
- Designed and implemented missing functionalities in C/C++ using Visual Studio Code, with a focus on modularity and maintainability.
- Accelerated team productivity by mentoring new team members and streamlining onboarding development processes, fostering collaboration, and providing ongoing technical guidance to ensure smooth integration and productivity.
- Acted as Tech Lead to preserve project continuity, interfacing directly with clients, leading technical discussions, and resolving five critical roadblocks that preserved project timeline, scope, and technical objectives.

ENEA Software Development Services

Senior Embedded Software Engineer

April 2021 - April 2022 (1 year)

Iaşi, Romania

- Strengthened code quality and maintainability through structured peer reviews and documentation.
- Aligned software architecture with mission-critical requirements, contributing to successful deployment in constrained environments.

- Improved system responsiveness and reduced latency in command handling by implementing efficient buffer management and interrupt-driven design, enhancing real-time performance.
- Utilized Xilinx SDK and worked on Zynq®-7000 SoC with dual-core ARM® Cortex™-A9 processors, optimizing performance for constrained environments.
- Engineered software modules to interface with the SpaceWire driver, enabling real-time processing of Telecommands sent from the On-Board Computer.
- Implemented a state machine to manage communication flow between the RendezVous Sensor Processing Unit and the On-Board Computer.
- Developed embedded software in C/C++ for space-grade platforms, focusing on reliability and fault tolerance.
- Prepared comprehensive technical documentation, including Interface specifications, Module descriptions, and Integration guidelines.
- Championed creation of a centralized knowledge base of 30+ software module descriptions, reducing integration errors by 20% and streamlining team communication across three teams.
- Collaborated with system architects and validation teams to ensure documentation accuracy and completeness, contributing to long-term maintainability and knowledge transfer.

RINF TECH
Software Development Consultant
June 2014 - August 2020 (6 years 3 months)
Sofia, Sofia City, Bulgaria

VISTEON Corporation
Senior Embedded Software Engineer (Contractor from SC RINF Outsourcing Solutions SRL)
June 2014 - April 2020 (5 years 11 months)
Sofia, Bulgaria

Projects : VS Daimler 2.0, MFA2 for Trucks and for Cars, MRA2 Cars
(Client: Daimler) – Instrument Cluster (IC)
- Led development of instrument cluster software in C/C++ for Daimler’s automotive platforms, improving feature reliability and user experience.
- Validated critical display components using CANoe simulation configurations and debugger tools for testing and validation of Maintenance Interval Display (WIA = WartungsIntervallAnzeige) component, reducing test cycle time by 20%.
- Delivered robust implementations of key features (e.g. Time and Date, Odometer, Speedometer, Tachometer, Language Setting, Outside Air

Temperature, Steering Wheel Switches, Coolant Temperature), contributing to successful OEM integration.

- Developed and executed Unit Tests using VectorCast, improving code reliability and reducing defects.
- Generated and integrated Franca Interface Definition Language (.fidl) files and implemented Inter-Process Communication (IPC) mechanisms to facilitate communication between Application Layer and HMI.
- Collaborated with cross-functional teams to ensure alignment with AUTOSAR RTE architecture and project specifications.
- Technologies and protocols: Autosar RTE, ANSI C, MULTI IDE, Visual C++ 6.0 IDE, PVCS, Interractive MID cluster, Inter-process communication, Renesas flashing tools for resources, Serena TeamTrack, Jenkins, Rational Team Concert (RTC), CANoe, CANalyzer

CONTINENTAL Automotive Romania SRL
Embedded Software Engineer
August 2010 - June 2014 (3 years 11 months)

- Implemented customer-specific functionalities in C/C++ based on detailed specifications and design models.
- Improved diagnostic reliability across 5+ OEM platforms by configuring faults (DTC), freeze frames, failure reactions, and inhibition mechanisms enhancing system robustness.
- Set up and validated CAN channels and diagnostic instances, ensuring robust error management.
- Ensured software robustness through thorough unit and integration testing, reducing field failures, and documenting the results in quality assurance reports.
- Collaborated with function developers to resolve specification discrepancies and improve implementation accuracy.
- Worked on multiple customer projects including Ford, Renault, Volkswagen, Porsche and Audi as part of the Competence Center Iasi Team.
- Technologies and protocols: CAN communication, CCP, XCP, OSEK RTOS, V-cycle model, Powertrain systems, ISO26262.

Education

Faculty of Electronics, Telecommunications and Information Technology, Iasi, Romania
Master’s Degree, Telecommunication Networks · (2009 - 2011)

Faculty of Economy and Business Administration, "Al.I.Cuza" University, Iasi, Romania
Bachelor’s Degree, Economics and Management · (2007 - 2010)

Faculty of Automatic Control and Computer Engineering, "Gh.Asachi" University, Iasi, Romania
Bachelor’s Degree, Information Technology · (2005 - 2009)

"Mihail Kogalniceanu" Highschool, Vaslui, Romania
High School, Mathematics and Computer Science · (2001 - 2005)

"Stefan cel Mare" School no 5, Vaslui, Romania
Primary School · (1993 - 2001)